

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 12B.3

ENTHALPY CHANGES OF SOLUTION AND HYDRATION

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	12: Entropy and Energetics
SUB-TOPIC:	12B.3: Enthalpy Changes of Solution and Hydration
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	2 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 12B.3

Q1. Define the standard enthalpy of hydration of an ion.
[WCH14/01/Oct21/Q5(a)]

[2 Marks] WCH14/01

Q2. Calculate the enthalpy change of solution for magnesium chloride. Given: $\Delta_{\text{LEH}}(\text{MgCl}_2) = -2526$, $\Delta_{\text{hydH}}(\text{Mg}^{2+}) = -1921$, $\Delta_{\text{hydH}}(\text{Cl}^-) = -364$ (all in kJ mol^{-1}). [WCH14/01/Jan21/Q2(b)]

[3 Marks] WCH14/01

SECTION B: WORKED MARK SCHEME — 12B.3

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: Define the standard enthalpy of hydration of an ion. [WCH14/01/Oct21/Q5(a)] The enthalpy change when one mole of gaseous ions (1) dissolves in sufficient water to form an infinitely dilute solution (1) under standard conditions.	[2]
Q2	Q2: Calculate the enthalpy change of solution for magnesium chloride. Given: $\Delta_{\text{LEH}}(\text{MgCl}_2) = -2526$, $\Delta_{\text{hydH}}(\text{Mg}^{2+}) = -1921$, $\Delta_{\text{hydH}}(\text{Cl}^-) = -364$ (all in kJ mol^{-1}). [WCH14/01/Jan21/Q2(b)] $\Delta_{\text{solH}} = \Delta_{\text{hydH}}(\text{cation}) + 2 \times \Delta_{\text{hydH}}(\text{anion}) - \Delta_{\text{LEH}}(1) = -1921 + 2(-364) - (-2526) (1) = -2649 + 2526 = -123 \text{ kJ mol}^{-1} (1)$	[3]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 12B.3**FAQ #1: Why do students make mistakes calculating $\Delta_{\text{sol}} H$?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Why do students make mistakes calculating $\Delta_{\text{sol}} H$?

MODEL ANSWER / EXAMINER EXPECTATION:

They forget to multiply the hydration enthalpy of the anion (e.g. Cl^-) by 2 when there are two anions in the empirical formula (e.g. MgCl_2). Always check the stoichiometry.

FAQ #2: How does solubility relate to these enthalpy changes?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

How does solubility relate to these enthalpy changes?

MODEL ANSWER / EXAMINER EXPECTATION:

If $\Delta_{\text{sol}} H$ is highly endothermic, the substance is likely insoluble. For a substance to be soluble, $\Delta_{\text{sol}} H$ should be exothermic or only slightly endothermic (so that ΔS_{total} remains positive).